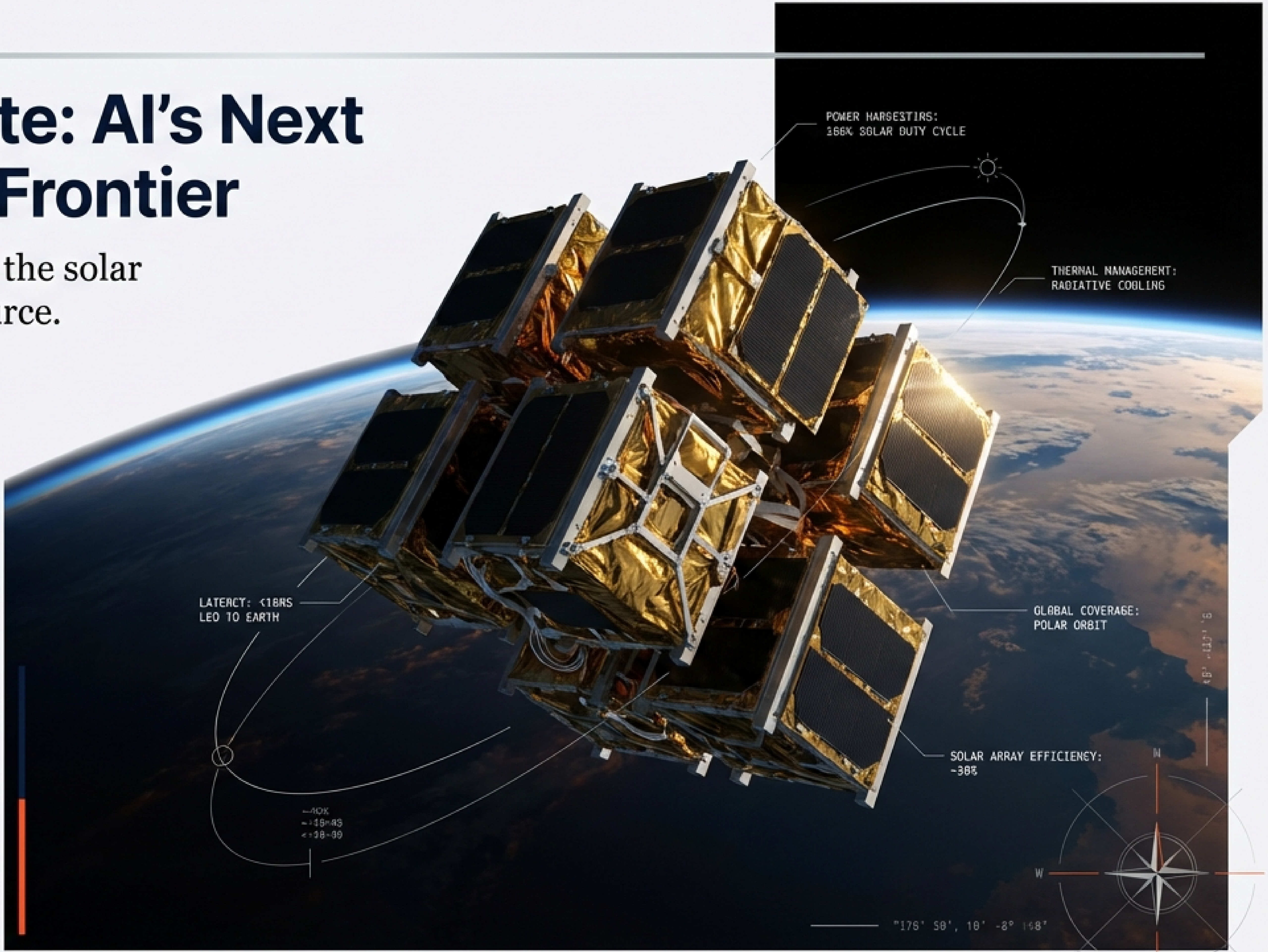
# Orbital Compute: AI's Next Infrastructure Frontier

Shifting the training layer to the solar system's ultimate energy source.

## SITREP:

As terrestrial AI demand outpaces the capacity of Earth's power grids, we investigate the engineering feasibility of migrating data centers to Low Earth Orbit (LEO).

Synthesis of industry analysis (New Atlas) and technical feasibility research (Google Project Suncatcher).




NotebookLM

# The Terrestrial Ceiling: Why Earth Can’t Keep Up

## The Power Crunch

Training frontier models (GPT-5, Claude 4) requires running thousands of GPUs simultaneously for weeks. This 24/7 load places unprecedented strain on local electrical grids and requires gigaliters of fresh water for evaporative cooling.

## The Resource Gap

Terrestrial data centers are increasingly constrained by:
— Land availability & Zoning
— High-voltage transmission limits
— Local resource competition (Water/Power)

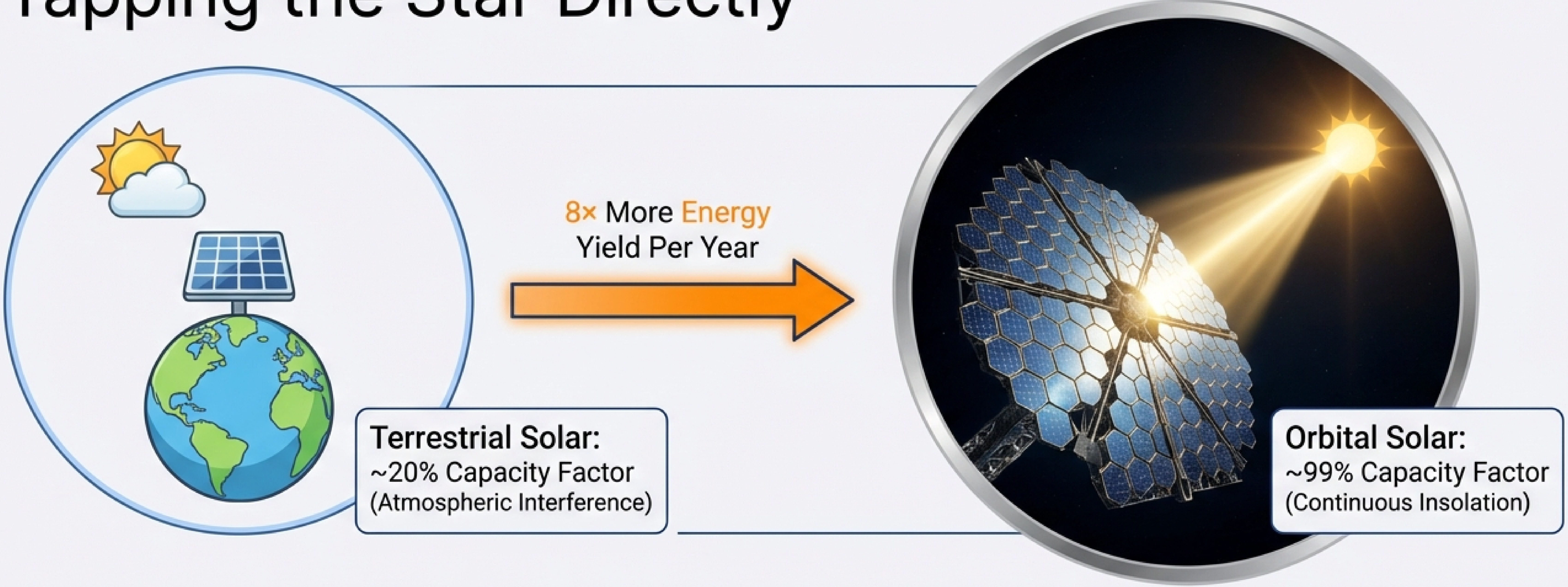
## The Divergence

Despite Google reducing Gemini query energy by 33×, total demand for AI services is outpacing efficiency gains. The bottleneck for AI scaling is no longer silicon; it is energy.

CONSTRAINT:

**PHYSICAL GRID CAPACITY**

# Tapping the Star Directly




## The Physics

The Sun emits 3.86 × 10^26 Watts. This is 100 trillion times humanity's total electricity production.

## The Pivot

Previous concepts focused on Space-Based Solar Power (beaming energy *down*). The new paradigm is Orbital Compute (moving the GPUs *up*).

## Key Players

— Google (Project Suncatcher)
— SpaceX / xAI (Projected merger)
— Starcloud (Training LLMs in orbit)
— Aetherflux

# Architecture: The Modular Swarm

## No Monoliths

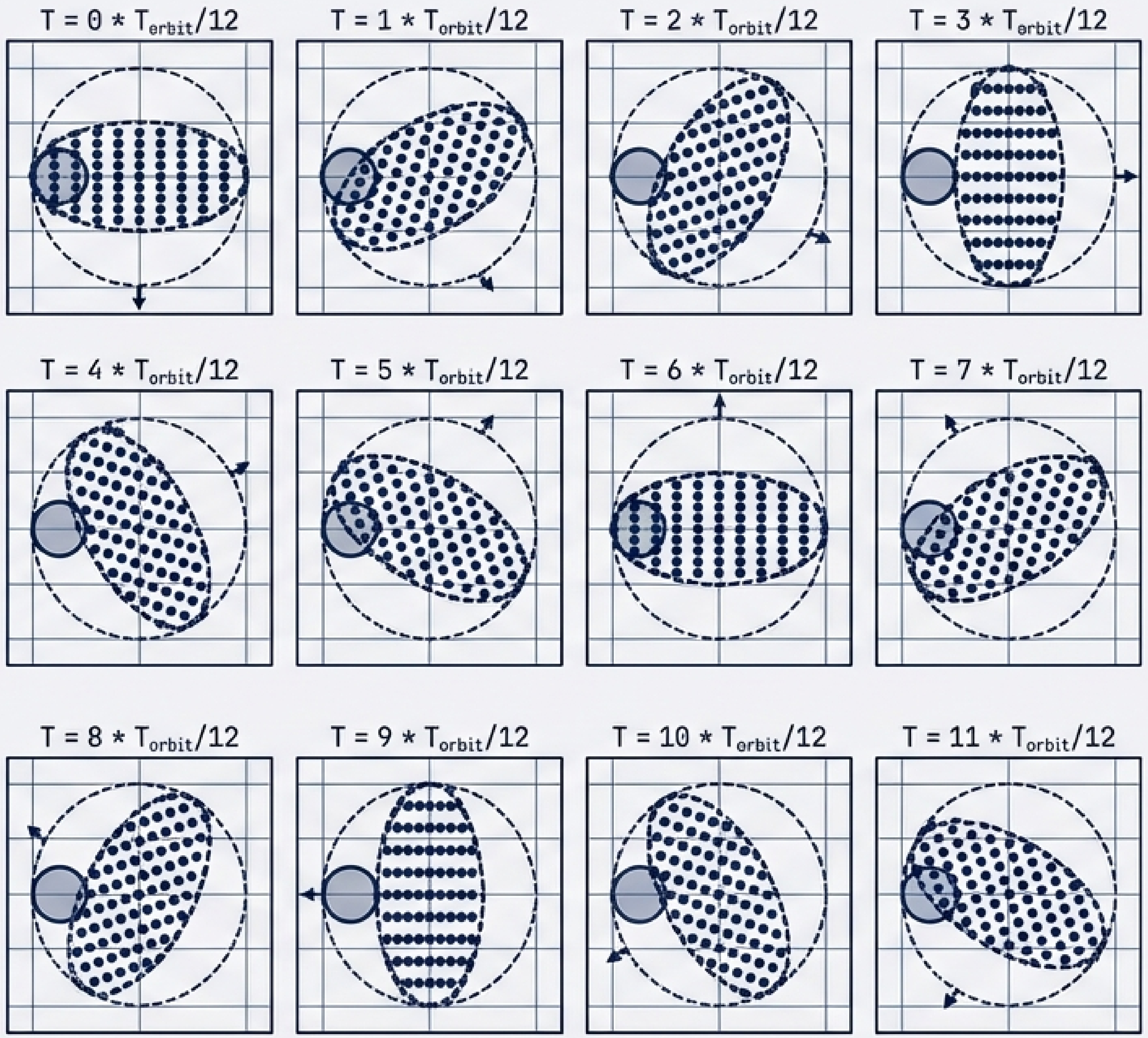
Building massive single structures in space is risky. The solution is a distributed supercomputer in a vacuum.

## The Suncatcher Proposal

— Cluster Size: 81 independent satellites
— Formation Radius: 1km
— Payload: Google Tensor Processing Units (TPUs)

## Orbit Profile: Dawn-Dusk Sun-Synchronous

The cluster rides the terminator line between day and night. This ensures permanent sunlight exposure for power while maintaining line-of-sight for inter-satellite links.



# Solving Connectivity: Proximity is Power

### The Solution

By flying satellites close (10m–1km), we can use COTS Dense Wavelength Division Multiplexing (DWDM).

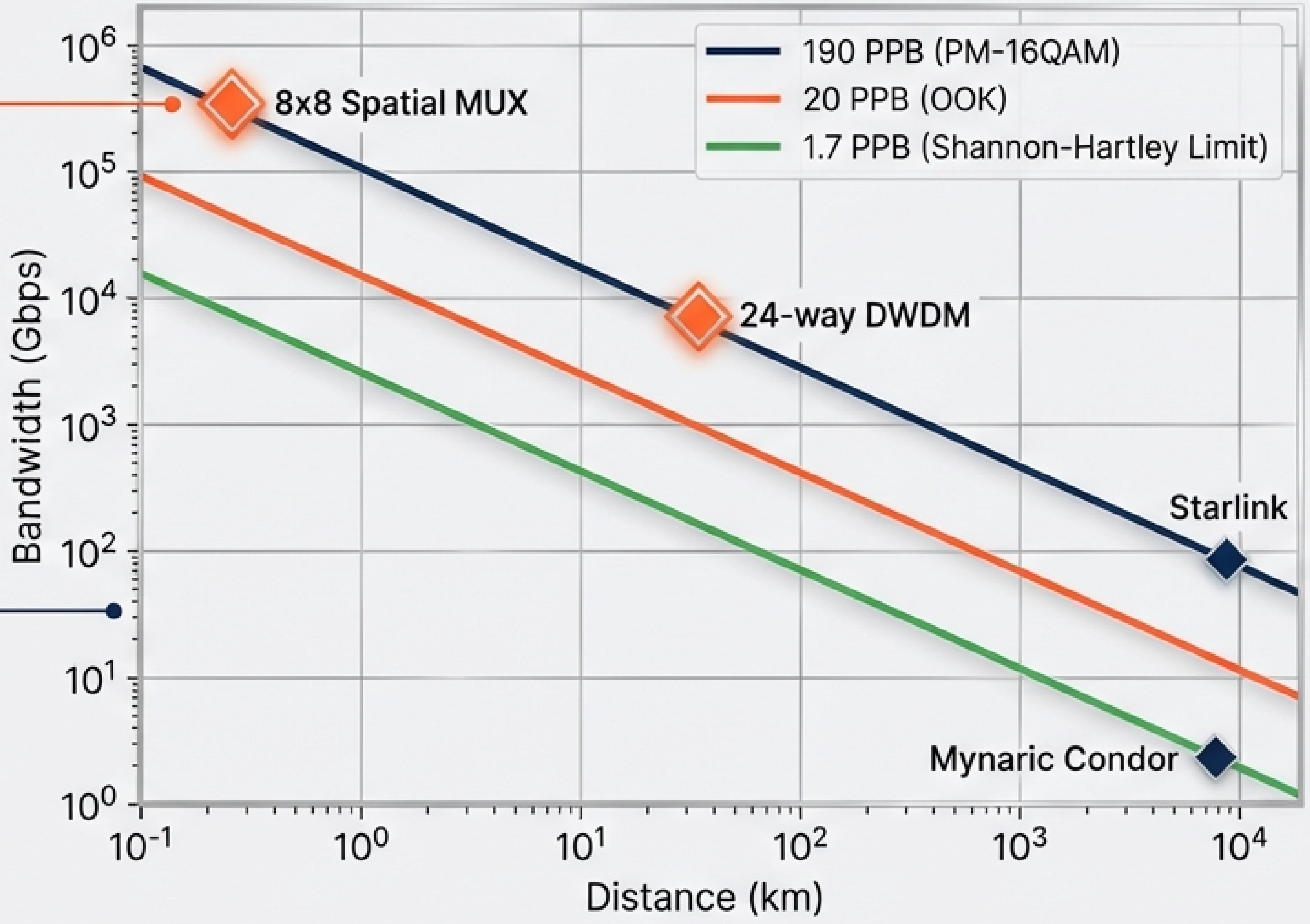
**Result: 10Tbps+ per link.** Matching terrestrial fiber speeds in a vacuum.

### The Physics

Received optical power scales with the inverse square of distance (1/d²).

$$P_{rx} \propto 1/d^2$$

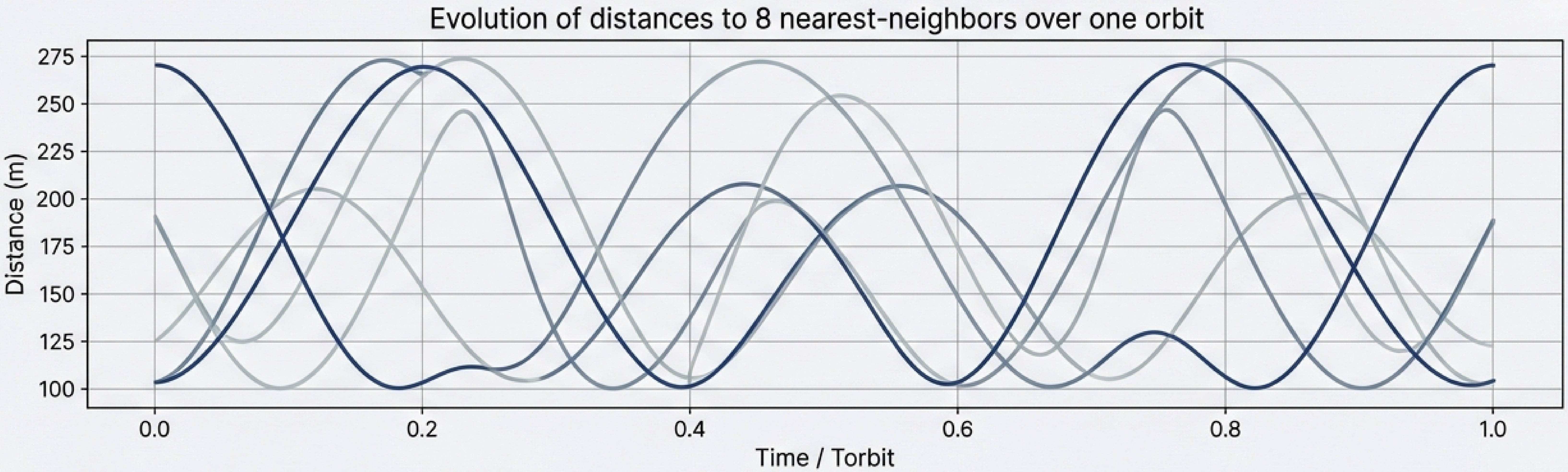
Bandwidth vs. Distance for a 5W, 10cm Telescope ISL




### The Challenge

Standard optical links (Starlink) operate at ~100Gbps over thousands of km. AI clusters need Terabits.

# Orbital Ballet: Dynamics and Control




### Formation Flight

The cluster isn't static; it breathes. As shown above, next-nearest-neighbor distances oscillate between 100m and 200m over the course of a single orbit.

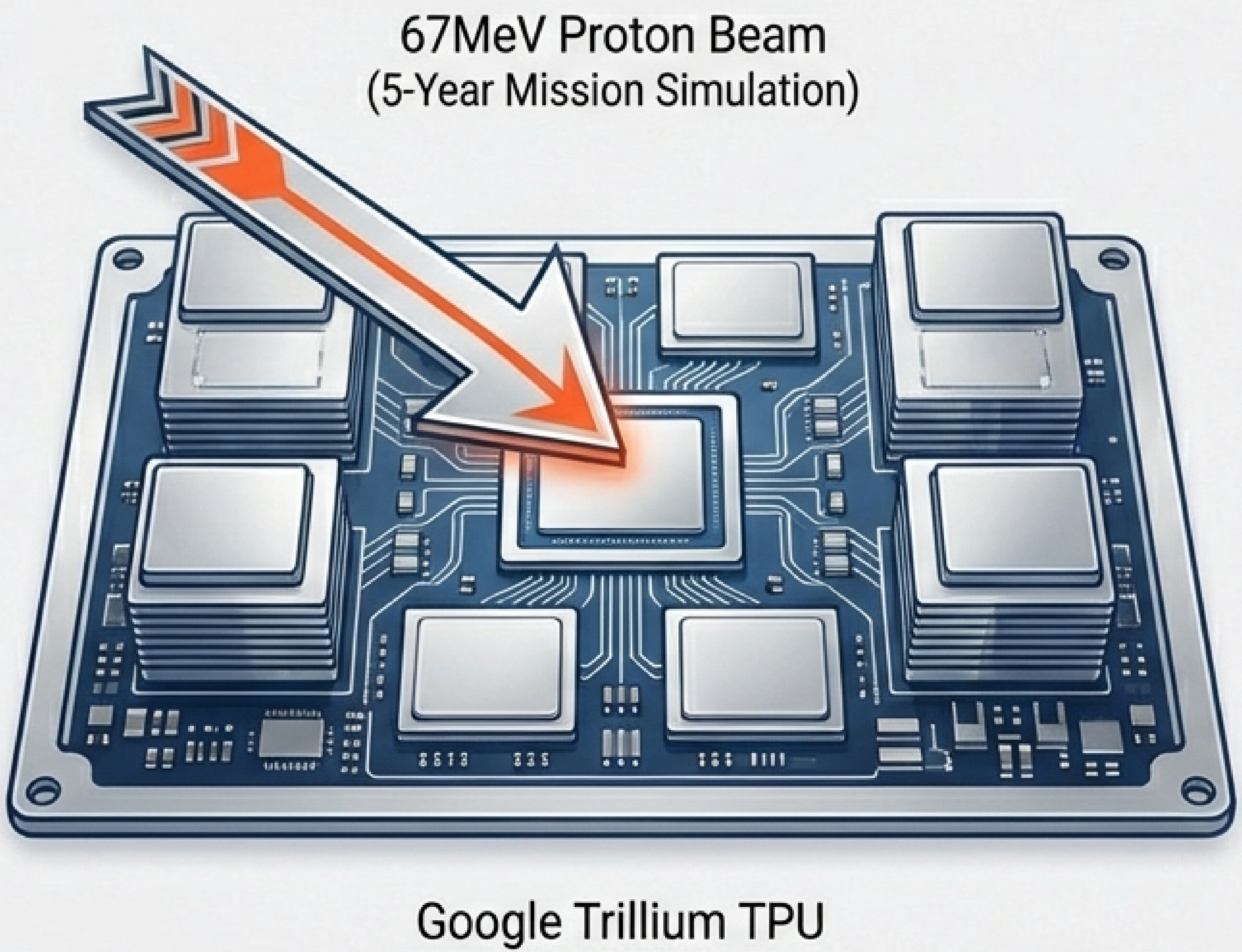
### The J2 Problem & ML Control

Earth is oblate (bulges at the equator), causing orbital drift. High-precision ML flight control models use backpropagation to predict this drift and automate station-keeping.

### Delta-V Requirements

Fuel costs are manageable. Station keeping requires only ~3 m/s/year per km of cluster radius.

# The Hardware Challenge: Will the Chips Survive?




### Total Ionizing Dose (TID)

Chips survived 5-year equivalent doses with no permanent failure.

### Single Event Effects (Bit-Flips)

High Bandwidth Memory (HBM) is sensitive to particle strikes.

### Mitigation Strategy

Mandatory Error Correction Code (ECC).

Error Rate: ~1 per 10 million inferences.

**Verdict:** Acceptable for Inference; Managed Risk for Training.

# The Economic Equation: Launch Cost Thresholds

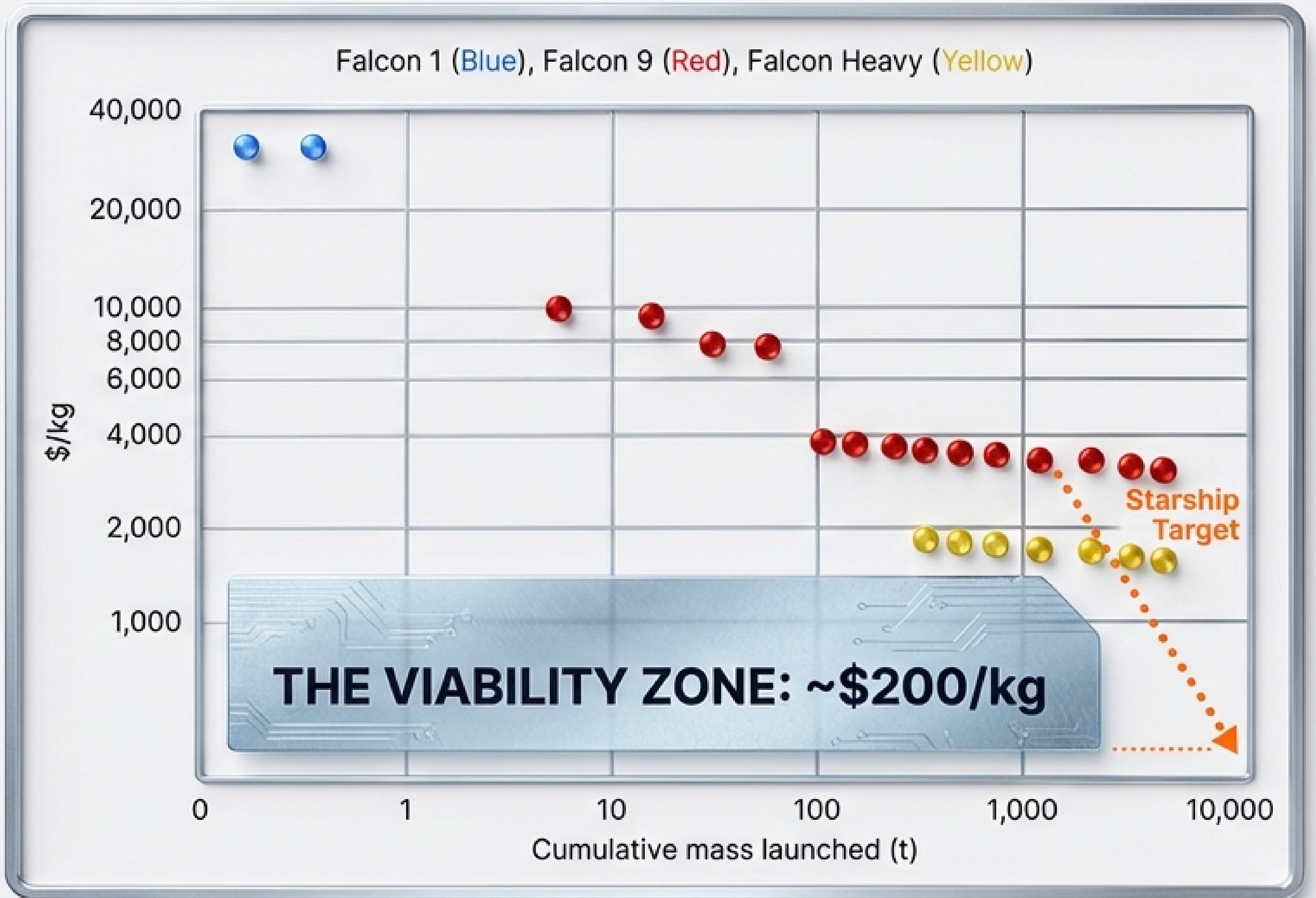
### The Metric: $/kg to LEO

The entire project lives or dies on launch costs.

**Current State (Falcon 9):**
`~$3,600/kg -> Power Cost ~$14,700/kW/y`

**Future Target (Starship @ Scale):**
`~$200/kg -> Power Cost ~$810/kW/y`

At $200/kg, space-based power achieves price parity with terrestrial data centers ($570–3,000/kW/y).



# Operational Hurdles & Risks

## Debris Environment

LEO contains 6,600 tons of junk and 14,000+ active satellites.

**Impact:** Dodging debris consumes fuel and interrupts compute cycles.

## Zero Maintenance

There are no technicians in space. A bad board cannot be swapped.

**Impact:** Massive redundancy (spare nodes) must be included in launch mass.


## Thermal Management

Vacuum is a perfect insulator; you cannot use fans.

**Impact:** Heat must be radiated away via massive surface areas, complicating structural design.

## Latency Physics

Speed of light is fixed.

**Impact:** Excellent for batch training (non-real-time), but creates lag for real-time inference (chatbots) compared to local edge compute.

# Roadmap: From Fantasy to Infrastructure

**Nov 2025**

Starcloud trains Shakespearean LLM in orbit.

**Next Year**

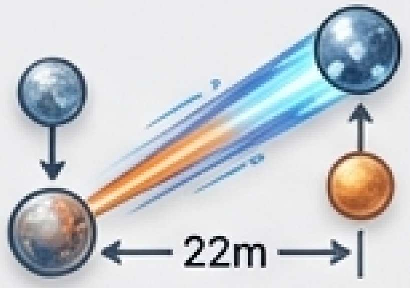
Google & Planet launch Suncatcher prototypes.

**2030s**

Projected Starship launch cost parity ($200/kg).

## The Verdict

- **The Physics work:** Proximity = Bandwidth.




- **The Hardware is validated:** TPUs survive radiation.

- **The Economics are converging:** Dependent on Starship execution.

**Conclusion:** We are witnessing the start of the first trillion-dollar infrastructure race in space. The question is not *if* compute moves to space, but *who* establishes the standard.